Mohab Kishawy

Ph.D. Candidate in Electrical and Computer Engineering, McMaster University | Available January 2027

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SUMMARY

Computer vision researcher working on physically grounded image restoration and 3D scene reconstruction, with first-author papers at ICPR 2026, the CVPR 2026 Workshops, and *Scientific Reports*, and four top-5 finishes in NTIRE challenges at CVPR (2024–2026). Seeking research scientist roles in computer vision and 3D vision from January 2027 (onsite or remote).

EDUCATION

Ph.D. in Electrical and Computer Engineering Sep 2023 – Dec 2026 (expected)
McMaster University Hamilton, ON, Canada

- Thesis: *Modelling the Physics of Degradation: Retinex-Based Restoration from 2D Images to 3D Scenes*
- Advisor: Prof. Jun Chen | GPA: 11.0/12.0 | Full scholarship

B.Sc. in Computer Engineering Sep 2018 – May 2023
Nile University Giza, Egypt

- Thesis: *Federated Learning System on Autonomous Vehicles for Lane Segmentation* (published in *Scientific Reports*)
- Advisor: Dr. Mohamed S. Darweesh | GPA: 3.7/4.0 | Full scholarship

RESEARCH EXPERIENCE

Computer Vision Researcher Sep 2023 – Present
McMaster University, advised by Prof. Jun Chen Hamilton, ON, Canada

- Physically grounded, Retinex-based restoration of ultra-high-definition images (Mamba, frequency-domain modelling) and of 3D scenes (3D Gaussian Splatting); released open-source code and pre-trained models.
- Led team entries in five NTIRE challenges at CVPR (2024, 2026) and competed in a sixth (2025), with four top-5 finishes.

Machine Learning Researcher Dec 2021 – Dec 2022
Helwan University, Electronics and Computer Engineering Lab Cairo, Egypt

- Automated assessment of students in online labs from mouse-tracking data, saving 32 hours of grading per term.

Research Assistant Feb 2021 – Aug 2021
Nile University, Wireless Intelligent Network Center (WINC) Giza, Egypt

- Plant-disease detection with a modified CNN; federated learning for lane segmentation on autonomous vehicles.

PUBLICATIONS

Earlier work appears under the name Mohab M. Eid. Full list with links: errorlogic1211.github.io/publications

Selected publications

Under review **M. Kishawy**, C. Tang, S. Shirani, J. Chen. **VeilGS: Solving Darkness as a Veil of 2D Basis Maps over Retinex 3DGS**. Manuscript available on request.

ICPR '26 **M. Kishawy**, A. A. Hussein, J. Chen. **RetinexDual: Retinex-Based Dual Nature Approach for Generalized Ultra-High-Definition Image Restoration**. International Conference on Pattern Recognition (ICPR), LNCS, Springer, pp. 90–105, 2026. Poster. [Paper] [arXiv] [Code] [Weights]

CVPRW '26 **M. Kishawy**, J. Chen. **RetinexDualV2: Physically-Grounded Dual Retinex for Generalized UHD Image Restoration**. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 1989–1998, 2026. Poster. [Paper] [arXiv]

Sci. Rep. '24 **M. M. Eid Kishawy**, M. T. Abd El-Hafez, R. Yousri, M. S. Darweesh. **Federated Learning System on Autonomous Vehicles for Lane Segmentation**. *Scientific Reports*, vol. 14, art. 25029, 2024. [Paper] [Code]

Other publications (NTIRE reports are co-authored as a participating team; our placement in brackets)

CVPRW '26 X. Li, Y. Jin, S. Yao, …, **M. Kishawy**, J. Chen, … (98 authors). *NTIRE 2026 The Second Challenge on Day and Night Raindrop Removal for Dual-Focused Images: Methods and Results*. CVPR Workshops, pp. 2054–2070, 2026. **[4th place]**

CVPRW '26 G. Ciubotariu, S. S. M. A, …, **M. Kishawy**, J. Chen, … (91 authors). *Low Light Image Enhancement Challenge at NTIRE 2026*. CVPR Workshops, pp. 1741–1753, 2026. **[5th place, Track 2: Joint Denoising and Low-Light Enhancement]**

CVPRW '26 R. P. Ancuti, A. Brateanu, …, **M. Kishawy**, … (89 authors). *NTIRE 2026 Nighttime Image Dehazing Challenge Report*. CVPR Workshops, pp. 1608–1625, 2026.

CVPRW '26 F.-A. Vasluianu, T. Seizinger, …, **M. Kishawy**, J. Chen, … (90 authors). *Advances in Single-Image Shadow Removal: Results from the NTIRE 2026 Challenge*. CVPR Workshops, pp. 2582–2595, 2026.

CVPRW '25 X. Liu, Z. Wu, F.-A. Vasluianu, …, **M. Kishawy**, J. Chen, … (105 authors). *NTIRE 2025 Challenge on Low Light Image Enhancement: Methods and Results*. CVPR Workshops, pp. 1196–1206, 2025. **[2nd place; highest PSNR]**

CVPRW '24 C. O. Ancuti, C. Ancuti, …, **M. Kishawy**, … (68 authors). *NTIRE 2024 Dense and Non-Homogeneous Dehazing Challenge Report*. CVPR Workshops, pp. 6453–6468, 2024. **[4th place]**

IEEE Access '22 A. M. Abd El-Haleem, **M. M. Eid**, M. M. Elmesalawy, H. A. H. Hosny. *A Generic AI-Based Technique for Assessing Student Performance in Conducting Online Virtual and Remote Controlled Laboratories*. *IEEE Access* 10:128046–128065, 2022. [Code] [Data]

CICN '22 H. A. Hassan, **M. M. Eid**, M. M. Elmesalawy, A. M. Abd El-Haleem. *A New Intelligent System for Evaluating and Assisting Students in Laboratory Learning Management System*. CICN, pp. 566–571, 2022. [Code] [Data]

IJAMC '21 M. M. El-Lamey, **M. M. Eid**, M. Gamal, N. M. Bishady, A. W. Mohamed. *Using Machine Learning Algorithms for Breast Cancer Diagnosis*. *Int. J. of Applied Metaheuristic Computing*, vol. 12, no. 4, pp. 117–137, 2021. Reprinted in *Research Anthology on Medical Informatics in Breast and Cervical Cancer*, IGI Global, 2022.

Springer '21 A. M. Ibrahim, N. E. M. Bishady, S. H. Elghalban, N. N. Mostafa, **M. M. Eid**, K. M. Abdelhamid. *Modelling and Analysis the Population Density Effect on the Infectiousness Rate of COVID-19 Novel Virus and the Mortality Rate Percentage Regarding Oxford's Stringency Index Model of Governmental Response in MATLAB*. In *Artificial Intelligence for COVID-19*, Springer, pp. 437–462, 2021.

NILES '20 A. M. Ibrahim, **M. M. Eid**, N. N. Mostafa, N. E. M. Bishady, S. H. Elghalban. *Modeling the Effect of Population Density on Controlling COVID-19 Initial Spread with the Use of MATLAB Numerical Methods and Stringency Index Model*. Novel Intelligent and Leading Emerging Sciences Conference (NILES), pp. 612–617, 2020.

HONORS AND AWARDS

2026 **4th Place**, NTIRE 2026 Day and Night Raindrop Removal Challenge, CVPR 2026
5th Place, NTIRE 2026 Joint Denoising and Low-Light Image Enhancement (JDLLIE) track, CVPR 2026
2025 **2nd Place**, NTIRE 2025 Low Light Image Enhancement Challenge, CVPR 2025 (highest PSNR)
2024 **4th Place**, NTIRE 2024 Dense and Non-Homogeneous Dehazing Challenge, CVPR 2024
2023 **1st Place**, Hacktrick AI Hackathon, Dell Technologies
2nd Place, AIC Competition for Arabic Text Summarization, Military Technical College, Egypt
Full Ph.D. Scholarship, McMaster University (2023–2026)
2022 **Best Solution Challenge Chapter (Global)**, Google Developer Student Clubs
Shield of Contribution, School of Engineering and Applied Sciences, Nile University
2021 **Dean's Honor List** (GPA above 3.75/4.00, Fall and Spring), Nile University
2018 **Full B.Sc. Scholarship**, Nile University (2018–2023)
2017 **3rd Place (Embedded Systems)**, Intel International Science and Engineering Fair (ISEF)
2015 **Bronze Medal**, Egyptian Olympiad in Informatics

TEACHING AND SERVICE

Teaching Assistant, McMaster University Sep 2023 – Present

COMPENG 2SH4 Principles of Programming (Fall 2025, Fall 2026) | COMPENG 3SM4 Algorithm Design and Analysis (Winter 2024, 2025, 2026) | ELECENG 3TQ3 Advanced Probability and Random Processes (Fall 2024) | COMPENG 4DM4 Computer Architecture (Fall 2023)

Reviewer: AAAI Conference on Artificial Intelligence (AAAI 2025); *IEEE Access* (6 reviews, 2022–2023); *Entropy* (2026)

SKILLS

Programming	Python, C/C++, MATLAB, Java, Bash
Deep learning	PyTorch, CUDA, TensorFlow/Keras, Hugging Face, timm, BasicSR, mamba-ssm, W&B, multi-GPU training
Vision and 3D	Image restoration and enhancement, 3D Gaussian Splatting, NeRF, OpenCV, segmentation (U-Net), YOLO, SAM
Other	Federated learning (Flower), NVIDIA Jetson/TensorRT, Git, Linux, LaTeX
Languages	English (fluent), Arabic (native)

LEADERSHIP AND OUTREACH

At Nile University: **Lead**, Google Developer Student Clubs (2021–2022; 500+ students, workshops and hackathons) | **Deep Learning Instructor**, Root Student Club (2021; 6-month bootcamp, 120+ students) | **Senator**, Student Union (2019–2021) | **ICPC Instructor** (2019–2020)